In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload()

df = pd.read_csv(next(iter(uploaded)))
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving california_housing_test.csv to california_housing_test (1).csv
```

In [ ]:
```
df.head()
```

Out[ ]:

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	hou
0	-122.05	37.37	27.0	3885.0	661.0	1537.0	
1	-118.30	34.26	43.0	1510.0	310.0	809.0	
2	-117.81	33.78	27.0	3589.0	507.0	1484.0	
3	-118.36	33.82	28.0	67.0	15.0	49.0	
4	-119.67	36.33	19.0	1241.0	244.0	850.0	

In [ ]:
```
# Calculate the correlation matrix for numerical columns
correlation_matrix = df.corr(numeric_only=True)

print('Correlation Matrix:')
display(correlation_matrix)
```

```
Correlation Matrix:
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms
longitude	1.000000	-0.925017	-0.064203	0.049865	0.070869
latitude	-0.925017	1.000000	-0.025143	-0.039632	-0.068245
housing_median_age	-0.064203	-0.025143	1.000000	-0.367850	-0.323154
total_rooms	0.049865	-0.039632	-0.367850	1.000000	0.937749
total_bedrooms	0.070869	-0.068245	-0.323154	0.937749	1.000000
population	0.111572	-0.117318	-0.299888	0.838867	0.856387
households	0.051062	-0.068296	-0.305171	0.914116	0.970758
median_income	-0.018701	-0.072363	-0.144315	0.221249	0.024025
median_house_value	-0.050662	-0.138428	0.091409	0.160427	0.082279

In [ ]:
```
column = "population"
print("Mean = ", df[column].mean())
print("Median = ", df[column].median())
print("Mode =", df[column].mode())
print("Minimum =", df[column].min())
print("Maximum =", df[column].max())
range_value = df[column].max() - df[column].min()
print("Range =", range_value)
print("Standard Deviation =", df[column].std())
print("Variance =", df[column].var())
print("Quantiles =", df[column].quantile())
```

```
Mean =  1402.7986666666666
Median =  1155.0
Mode = 0    870.0
Name: population, dtype: float64
Minimum = 5.0
Maximum = 11935.0
Range = 11930.0
Standard Deviation = 1030.5430124122422
Variance = 1062018.9004316987
Quantiles = 1155.0
```

Dataset Summary: Missing Values

In [ ]:
```
print('Missing values per column:')
display(df.isnull().sum())
```

```
Missing values per column:
```

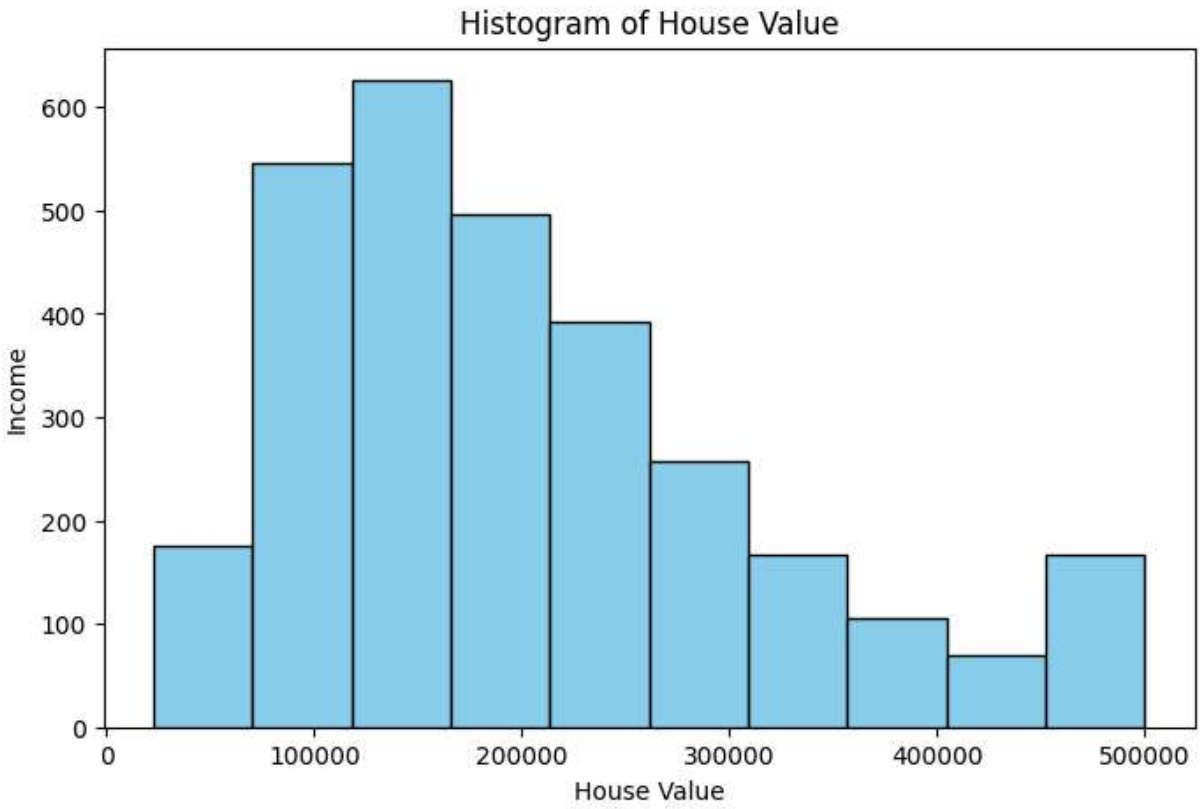
	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	0
population	0
households	0
median_income	0
median_house_value	0

dtype: int64

In [ ]:
```
column = "median_house_value"
plt.figure(figsize=(8,5))
plt.hist(df[column], bins=10, color='skyblue', edgecolor='black')

plt.title("Histogram of House Value")
plt.xlabel("House Value")
plt.ylabel("Income")

plt.show()
```



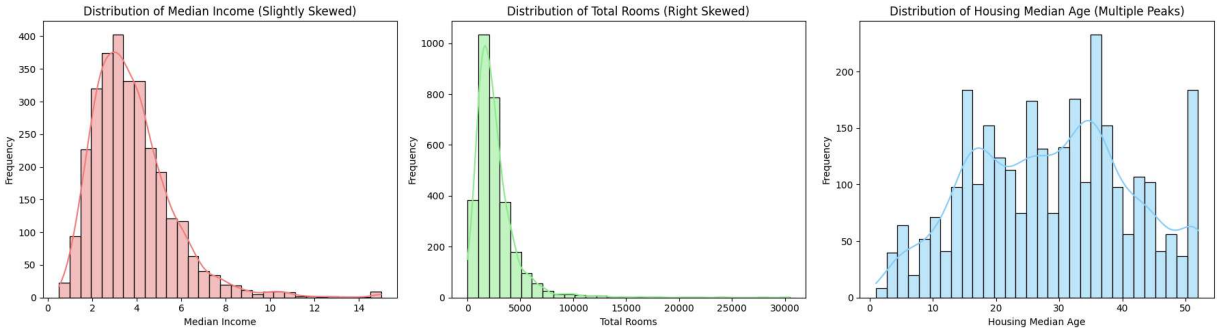
In [ ]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Histogram for a nearly normal/slightly skewed distribution (e.g., median_income)
sns.histplot(df['median_income'], bins=30, kde=True, ax=axes[0], color='lightcoral'
axes[0].set_title('Distribution of Median Income (Slightly Skewed)')
axes[0].set_xlabel('Median Income')
axes[0].set_ylabel('Frequency')

# Histogram for a highly skewed distribution (e.g., total_rooms)
sns.histplot(df['total_rooms'], bins=30, kde=True, ax=axes[1], color='lightgreen')
axes[1].set_title('Distribution of Total Rooms (Right Skewed)')
axes[1].set_xlabel('Total Rooms')
axes[1].set_ylabel('Frequency')

# Histogram showing frequency peaks (e.g., housing_median_age)
sns.histplot(df['housing_median_age'], bins=30, kde=True, ax=axes[2], color='lights
axes[2].set_title('Distribution of Housing Median Age (Multiple Peaks)')
axes[2].set_xlabel('Housing Median Age')
axes[2].set_ylabel('Frequency')

plt.tight_layout()
plt.show()
```

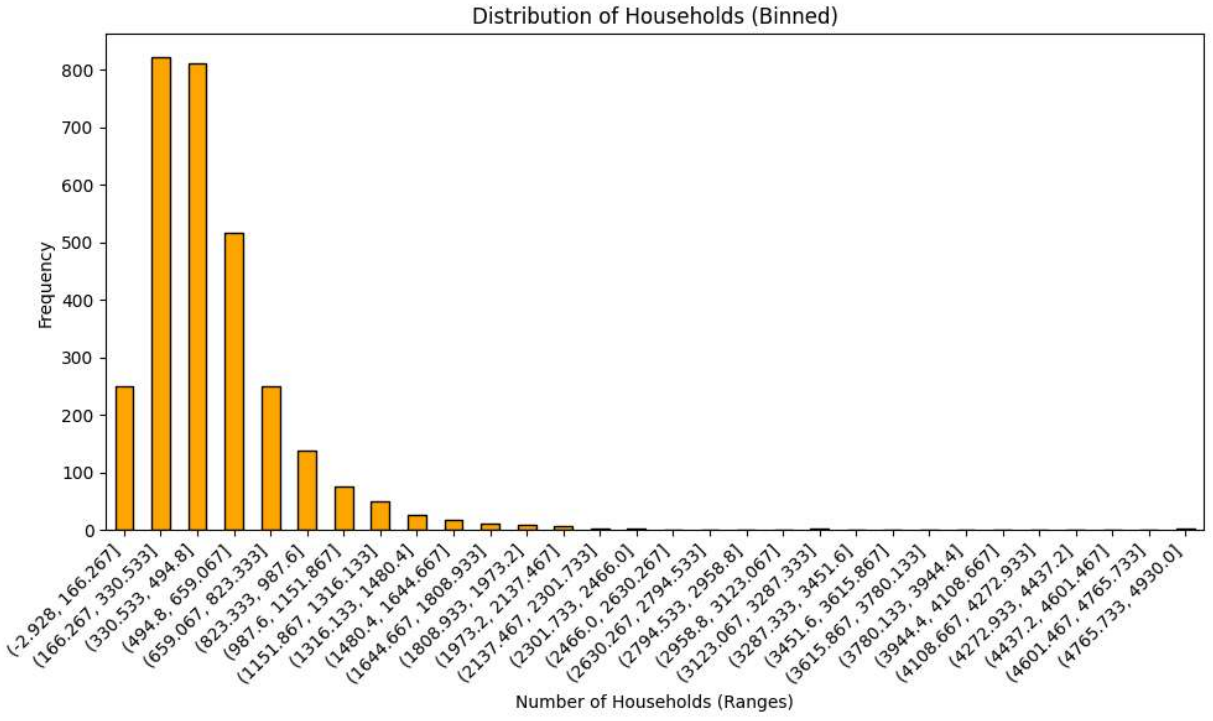


In [ ]:
```
column = "households"
plt.figure(figsize=(10,6))

# bins to show up as household data
bins = pd.cut(df[column], bins=30) # can adjust the no. of bins

# plotting the value counts of these bins as a bar chart
bins.value_counts().sort_index().plot(kind='bar', color='orange', edgecolor='black'

plt.title("Distribution of Households (Binned)")
plt.xlabel("Number of Households (Ranges)")
plt.ylabel("Frequency")
# xticks so the names show up easily, ha='right' + tight_layout so nothing overlaps
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

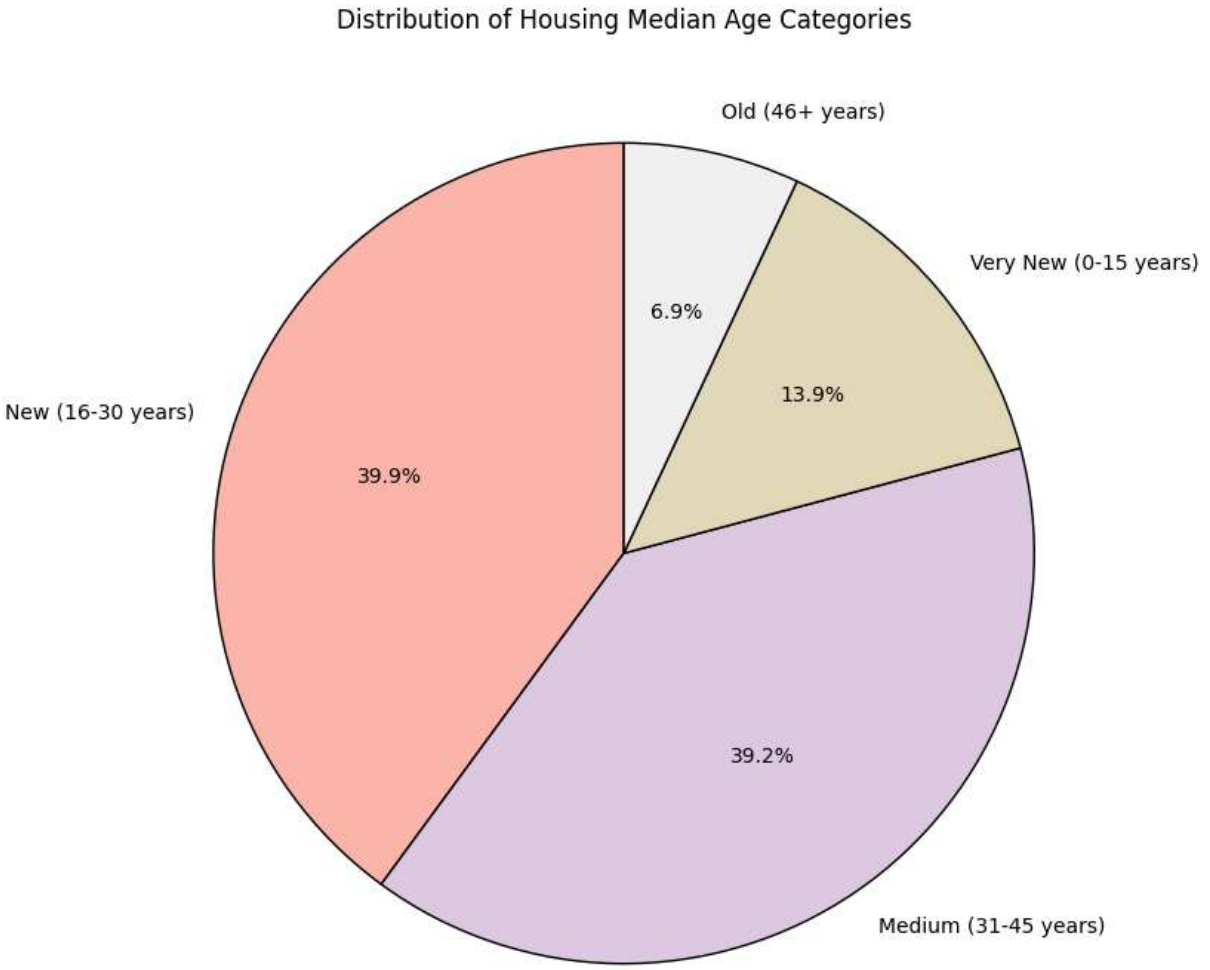


In [ ]:
```
column = "housing_median_age"
plt.figure(figsize=(9, 9))

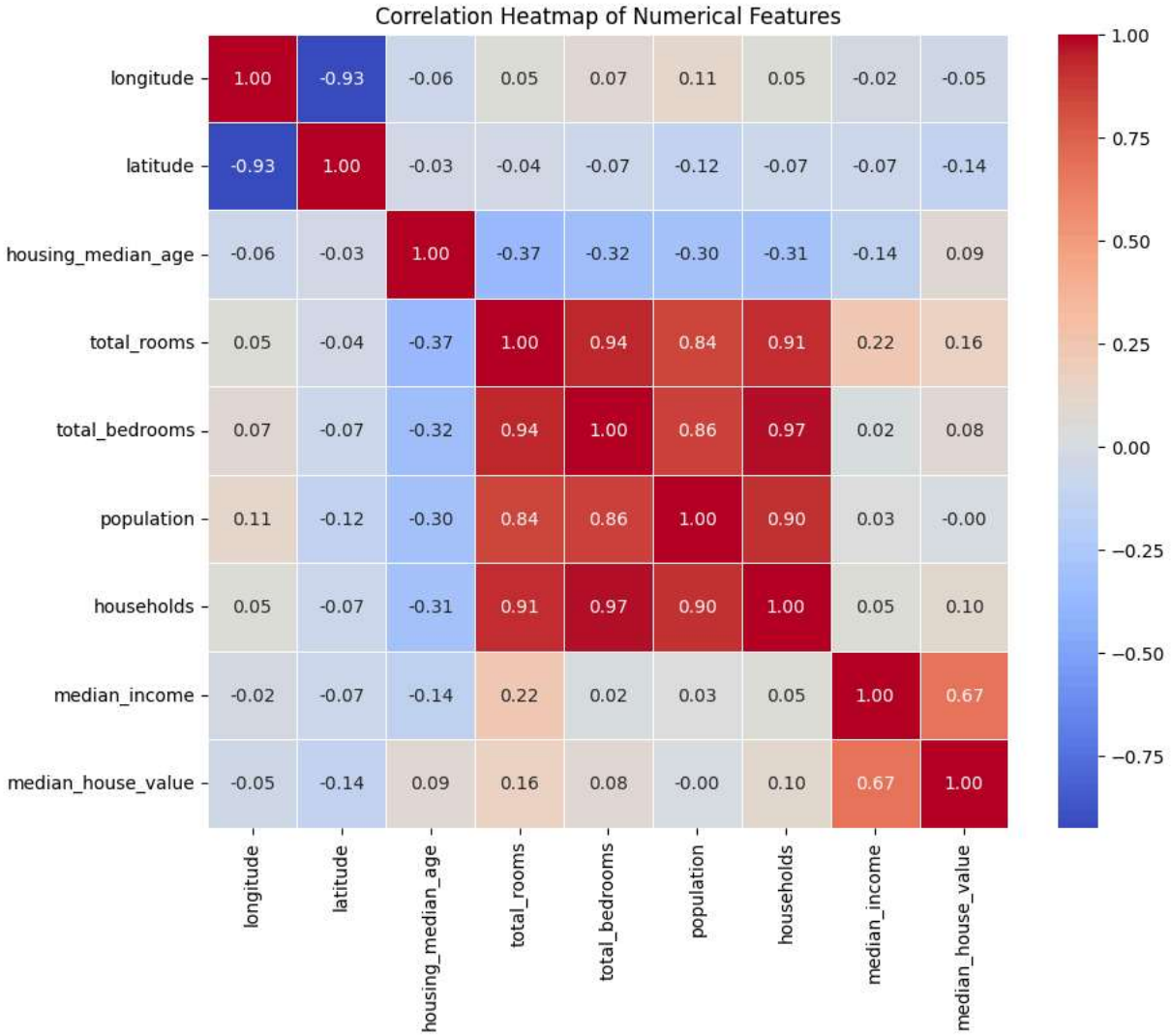
# categories for housing_median_age
bins = pd.cut(df[column], bins=[0, 15, 30, 45, df[column].max()],
              labels=['Very New (0-15 years)', 'New (16-30 years)', 'Medium (31-45
```

```
# value counts of these categories as a pie chart
bins.value_counts().plot(kind='pie', autopct='%1.1f%%', startangle=90, cmap='Pastel

plt.title("Distribution of Housing Median Age Categories")
plt.ylabel("") # hide the y-label for a cleaner pie chart
plt.show()
```

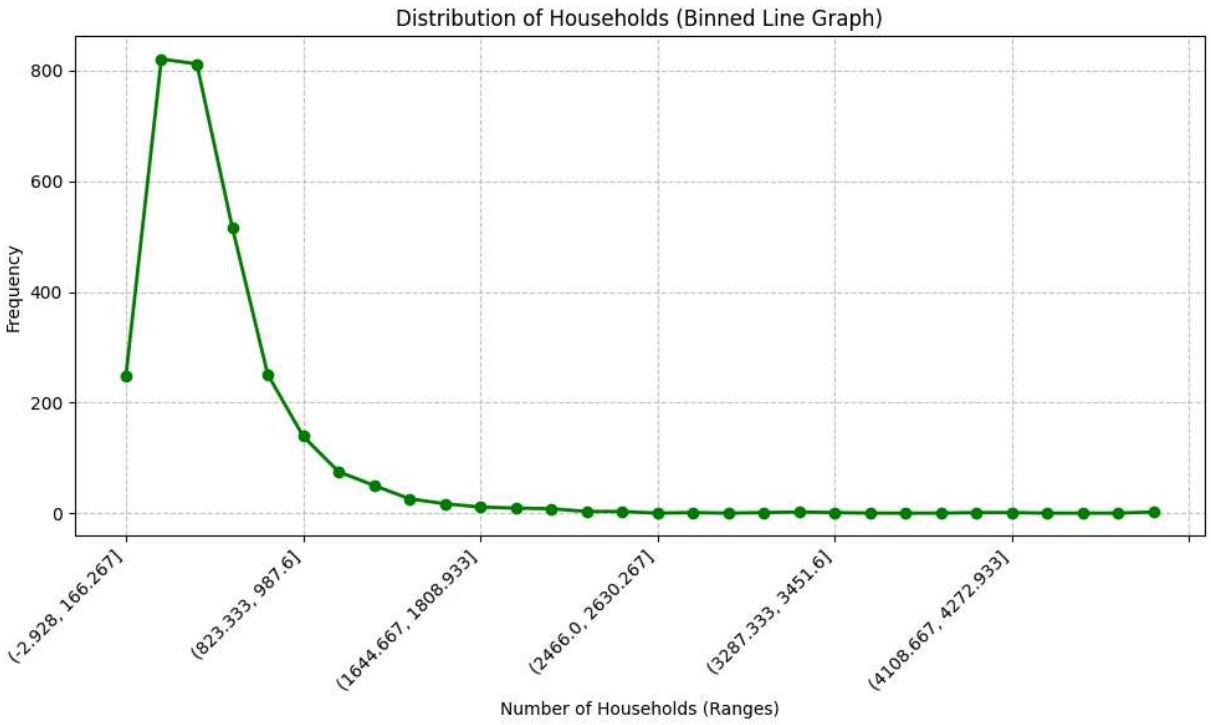


In [ ]:
```
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=
plt.title('Correlation Heatmap of Numerical Features')
plt.show()
```



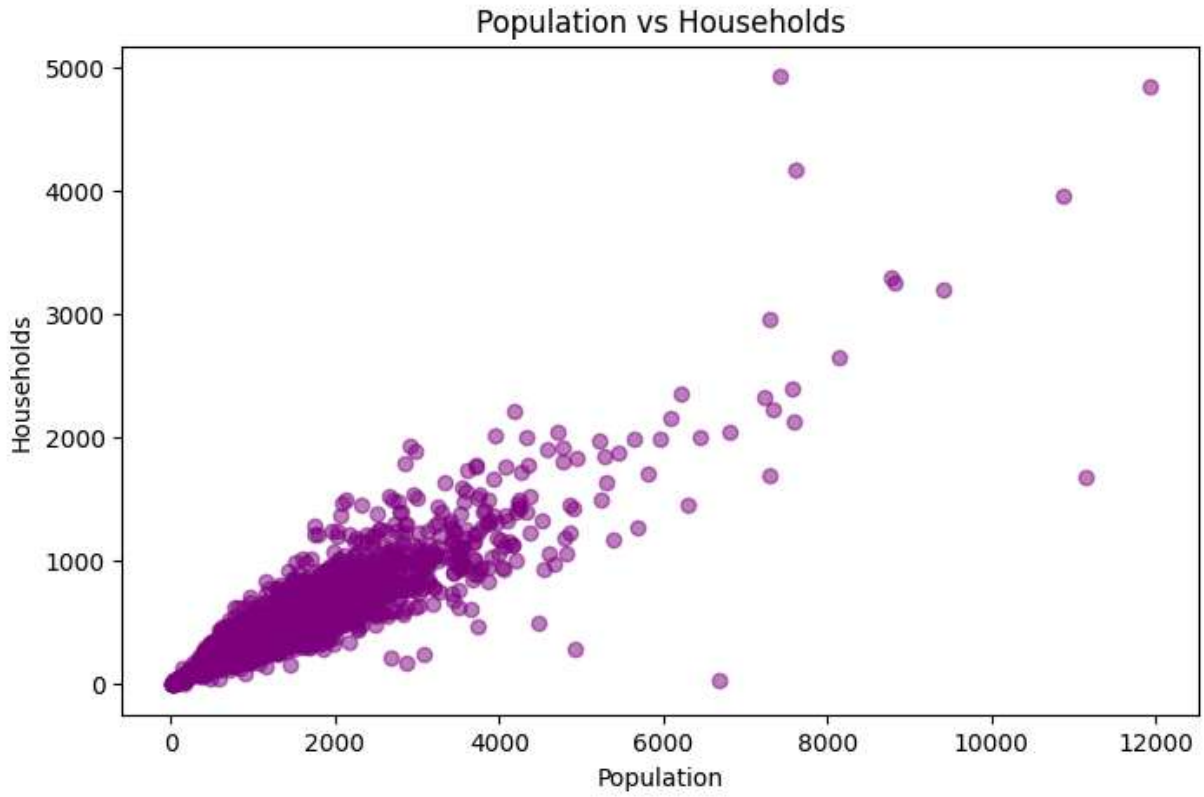
In [ ]:
```
column = "households"
plt.figure(figsize=(10,6))
bins = pd.cut(df[column], bins=30)
binned_counts = bins.value_counts().sort_index()
binned_counts.plot(kind='line', marker='o', color='green', linewidth=2)

plt.title("Distribution of Households (Binned Line Graph)")
plt.xlabel("Number of Households (Ranges)")
plt.ylabel("Frequency")
plt.xticks(rotation=45, ha='right')
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



In [ ]:
```
plt.figure(figsize=(8,5))
plt.scatter(df['population'], df['households'], color = 'purple', alpha=0.5)
plt.title('Population vs Households')
plt.xlabel('Population')
plt.ylabel('Households')

plt.show()
```



Conclusion from Data Exploration

Based on the analysis performed:

Missing Values:

- The analysis confirmed that there are **no missing values** in any of the columns. This means the dataset is clean in terms of missing data and doesn't require imputation or removal of rows for this specific issue.

Skewness and Normalization:

- **Right-skewed distributions:** Features like `total_rooms` and `population` are highly right-skewed. This suggests that there are many instances with lower values and fewer instances with very high values. For many machine learning models, transforming these features (e.g., using log transformation or square root transformation) to achieve a more normal distribution can improve model performance.
- **Slightly skewed/normal-like:** `median_income` appears to be more normally distributed with a slight right skew.
- **Frequency Peaks:** `housing_median_age` shows multiple peaks, indicating different phases of construction or housing development.

Correlations:

- **Highly Correlated Variables (Positive):**
  - `total_rooms` and `total_bedrooms` are highly positively correlated, which is expected.
  - `total_rooms`, `population`, and `households` also show strong positive correlations with each other.
  - `median_income` shows a moderate to strong positive correlation with `median_house_value`.
- **Negative Correlations:**
  - `longitude` and `latitude` show a strong negative correlation, reflecting the geographical distribution where increasing longitude generally means decreasing latitude in California's coastal regions.
  - `latitude` also shows a negative correlation with `median_house_value` (possibly indicating that house values decrease as you move north from highly desirable southern California areas).
- **Weak Correlations:**
  - `housing_median_age` generally has weaker correlations with most other features.

Feature Selection Suitability:

- **Highly correlated features:** For features like `total_rooms` and `total_bedrooms`, if one is a strong predictor, the other might be redundant. In some models, one might be selected over the other or a combined feature could be created to avoid multicollinearity issues.
- **`median_income` and `median_house_value`:** Given their strong positive correlation, `median_income` is a very suitable feature for predicting `median_house_value`.
- **Location features (`longitude`, `latitude`):** These are important and their strong negative correlation with each other, as well as their correlations with other features, indicate they are crucial for geographical context in predictions.

Overall, the data exhibits clear patterns of skewness in several features, suggesting potential transformations for modeling. Strong correlations exist between several housing-related attributes and with the target variable `median_house_value`, making them good candidates for predictive models.

In [ ]:
```
!jupyter nbconvert --to html /content/SMLD_EXP1.ipynb
```